Personal & Professional Project

Engineering Student — Software & Applied Systems

Education and Scientific background

CPGE BACKGROUND | M1 ENGINEERING

General Scientific Education

General Baccalaureate with a strong focus on Mathematics and Physics, providing early exposure to formal reasoning, abstraction, and quantitative problem-solving.

CPGE – MPSI / MP (Lycée Pierre-de-Fermat)

Two years of intensive scientific training emphasizing mathematical rigor, theoretical depth, and problem-solving under constraints, with a strong focus on analysis, algebra, and physics.

Engineering School – ENSEEIHT

Engineering curriculum oriented toward applied mathematics, computer science, and systems, bridging rigorous theory with concrete technical applications.

What Shapes My Approach

Strength Training

Builds discipline and confidence through long-term progression. Provides a structured outlet during intense academic or project periods.

Running & Hiking

Develops endurance and persistence. Also serves as a way to disconnect from cognitive overload and maintain focus over time.

Creative Outlets

Reading, drawing, and cooking help maintain balance and bring alternative perspectives to technical problem-solving.

Industry Exposure

Industry Perspectives

Discussions with a software developer / scrum master highlighted the importance of soft skills, team dynamics, and work–life balance in real-world software product development.

Technical Diversity in Tech

Exchange with a Database Administrator (DBA) broadened my understanding of the tech ecosystem beyond development alone, revealing the variety of technical roles and system responsibilities involved.

Corporate & Career Insight

Discussion with a former developer now in a management role provided insight into career evolution in tech companies, organizational structures, and the long-term employment stability of the sector.

Professional Events

Attendance at a Berger-Levrault conference (Toulouse) and discussion with their Lead AI Researcher reinforced the idea that software engineering remains a durable and evolving profession, even in the context of increasing automation and AI.

Professional Direction

Primary Path – Systems & Performance-Oriented Software Engineering

Focus on software engineering for **performance-critical and system-level applications**.

Areas of interest include **graphics and rendering systems**, **high-performance pipelines**, and **architecture of complex software systems**.

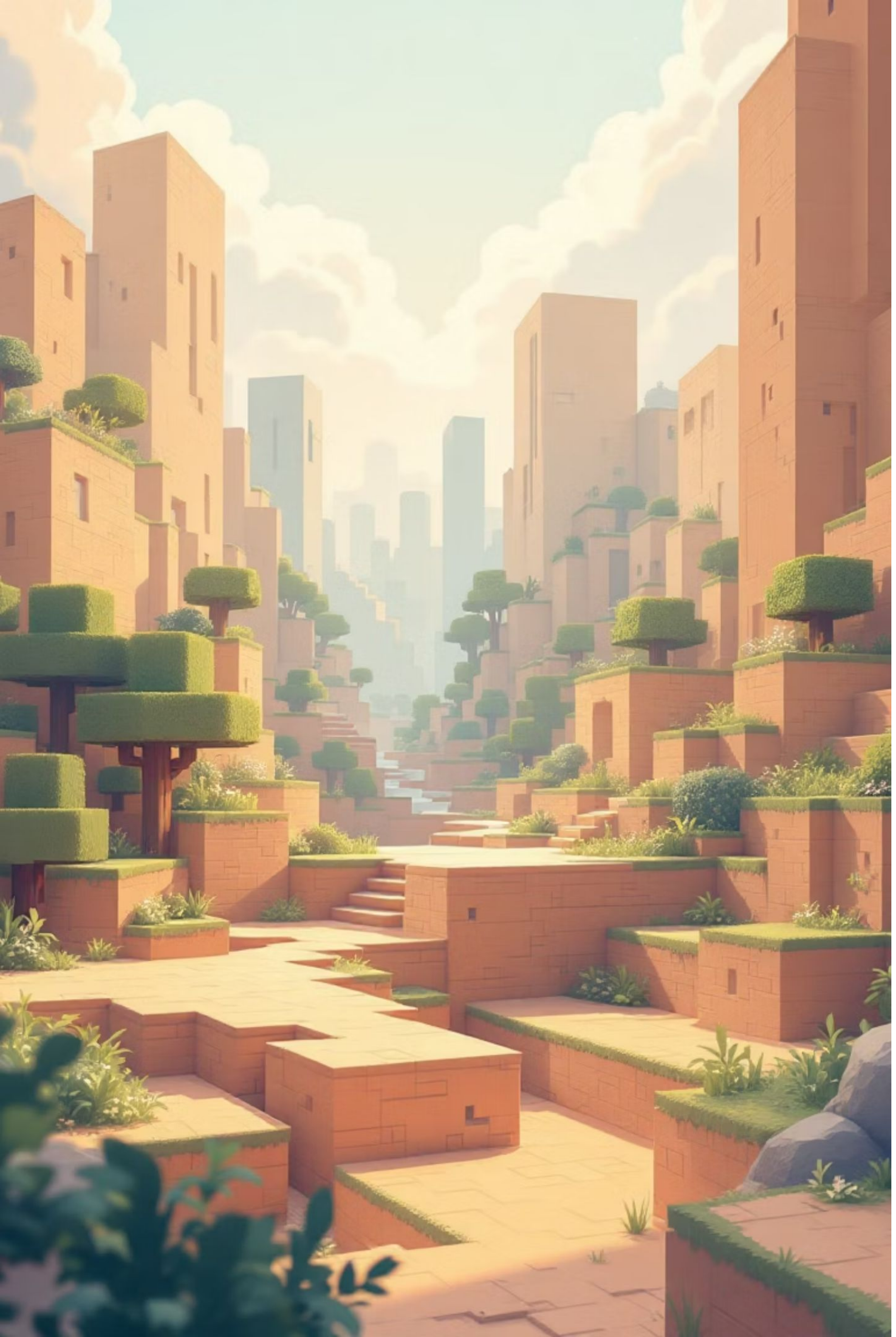
Secondary Path – Research-Oriented Track (Applied & Computational)

Applied mathematics, scientific computing, ML

A coherent secondary option oriented toward **scientific computing, applied mathematics, and imaging technologies.**

This interest is grounded in **concrete research exposure** (first-year internship in data analysis at INRAE) and reinforced by a **math- and image-processing–focused academic trajectory**.

Common Core : Both paths rely on the same foundations: **mathematical modeling, experimentation, performance analysis, and rigorous problem-solving**.



Core Technical Project

Voxel Rendering Engine

Genesis

Initially developed as a Java course project, the engine progressively evolved into a personal experimental platform to study **real-time rendering**, **memory management**, and **systems architecture** under performance constraints.

Technical Evolution

Chunk-based world representation with procedural terrain generation

OpenGL rendering pipeline optimisation for real-time performance

Multi-threaded terrain generation, decoupling computation from rendering to prevent frame drops

International Gap Year

Purpose-Driven Mobility

Planned year abroad designed to strengthen professional autonomy, international exposure, and practical experience in an English-speaking environment.

Two Possible Professional Formats

Technical internship in software engineering, aligned with systems- and performance-oriented development

Full-time professional work to achieve financial independence while discovering workplace culture and international living conditions

Target Locations

Canada or Australia—strong tech sectors, working holiday visa accessibility, cultural alignment with engineering career goals.

Professional & Personal Objectives

- Confront real constraints: employment conditions, housing, visas, budgeting

Work within **international teams and environments**

- Build autonomy before committing to a long-term specialization

Moving Forward

A Clear Technical Axis

Focused on **systems and performance-oriented software engineering**, supported by concrete technical projects and a strong theoretical background.

A Coherent Secondary Option

A research-oriented path in **scientific computing, applied mathematics, and imaging** remains a coherent extension of the same core skills.

A Structured Next Step

The upcoming **international gap year** is designed as an acceleration phase to gain autonomy, real-world exposure, and international professional experience.

Long-Term Objective

Build robust, high-impact technical systems by combining **rigorous engineering**, **experimentation**, and **continuous learning**.